

Subsystem — Altimeter MS5607

Sensor: MS5607-02BA03-50 — TE Connectivity

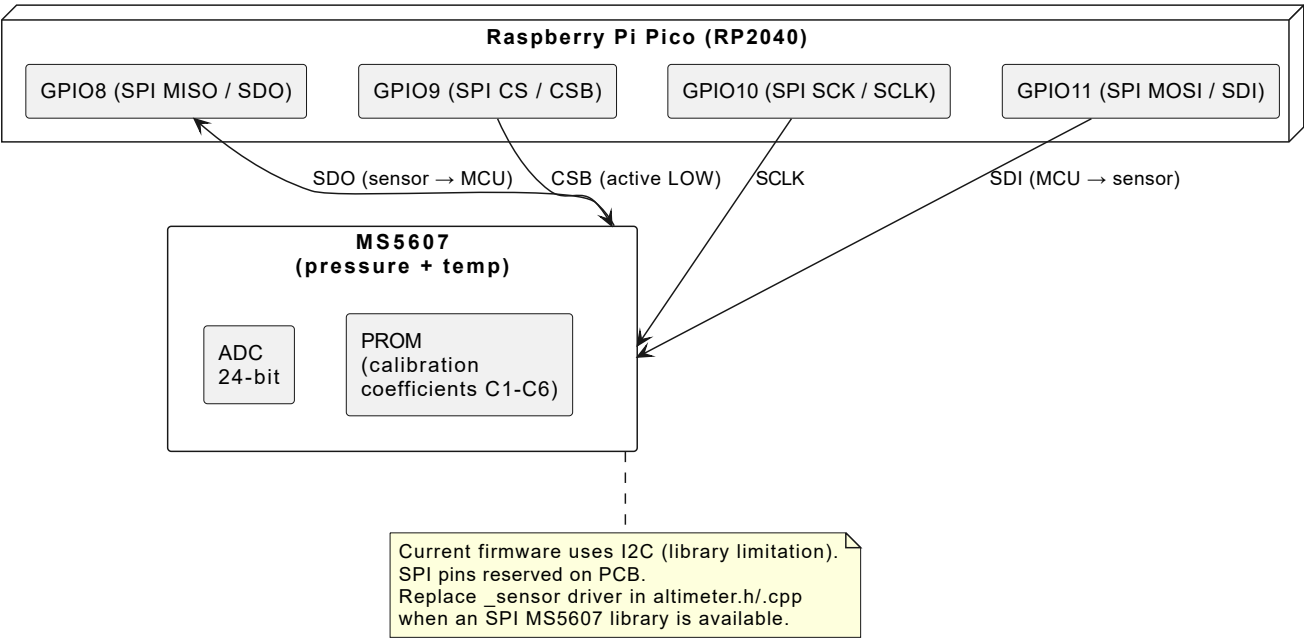
Measures: Pressure (10–1200 hPa) · Temperature (–40...+85 °C) · Altitude (m)

Hardware target: SPI on GPIO8–11

Current firmware: I2C via bundled MS5607 library ([Libnexamples/MS5607-master/](#))

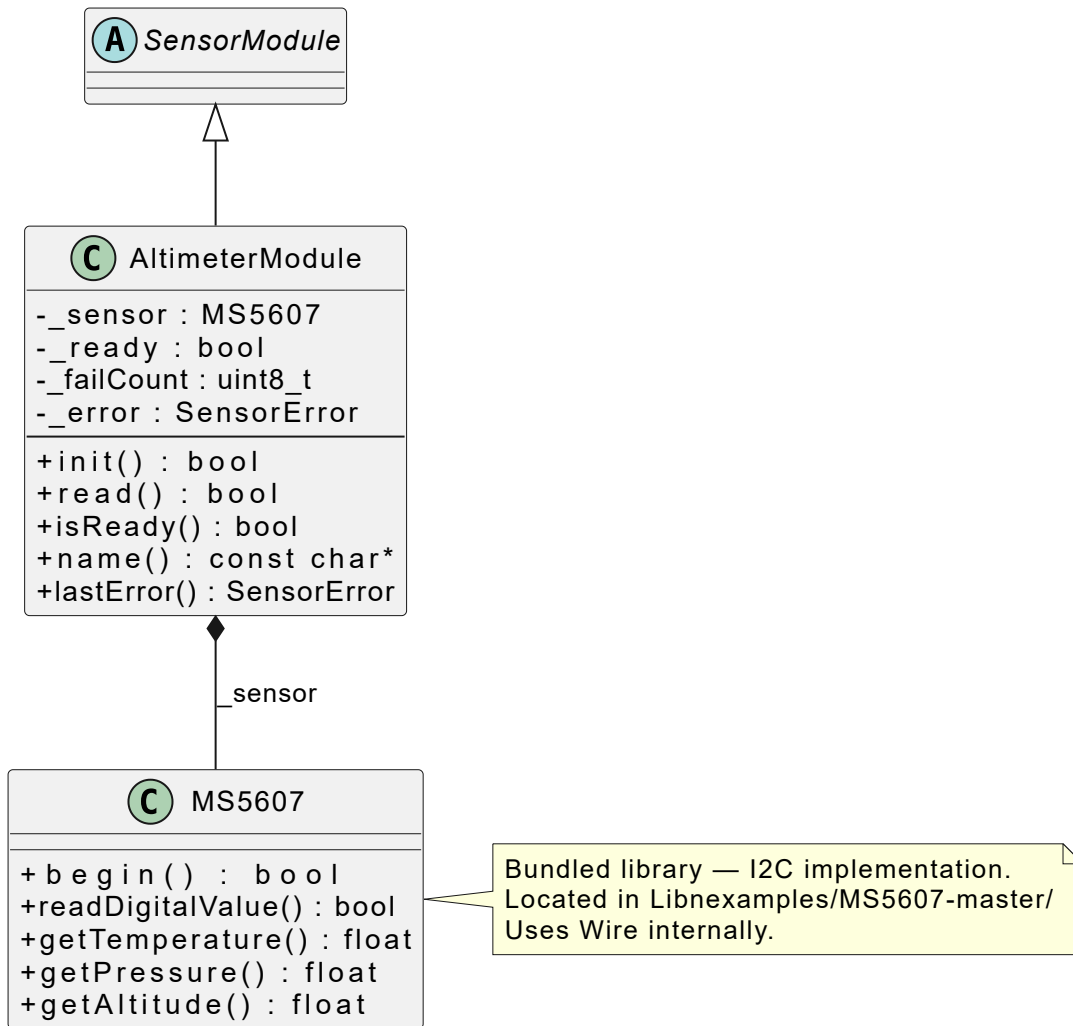
Files: [include/altimeter.h](#) · [src/altimeter.cpp](#)

1. Hardware Connections

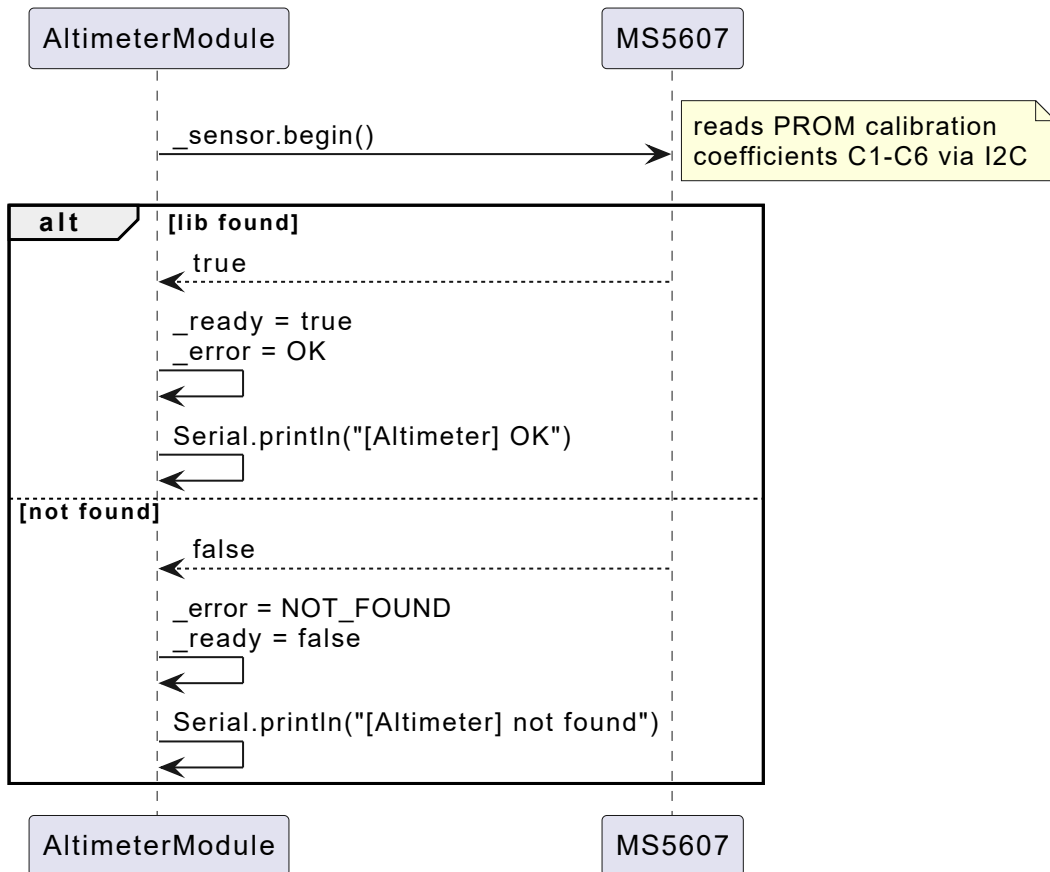


GPIO	SPI signal	MS5607 pin	Status
GPIO8	MISO	SDO	Reserved — not driven by current firmware
GPIO9	CS	CSB	Reserved — not driven by current firmware
GPIO10	SCK	SCLK	Reserved — not driven by current firmware
GPIO11	MOSI	SDI	Reserved — not driven by current firmware

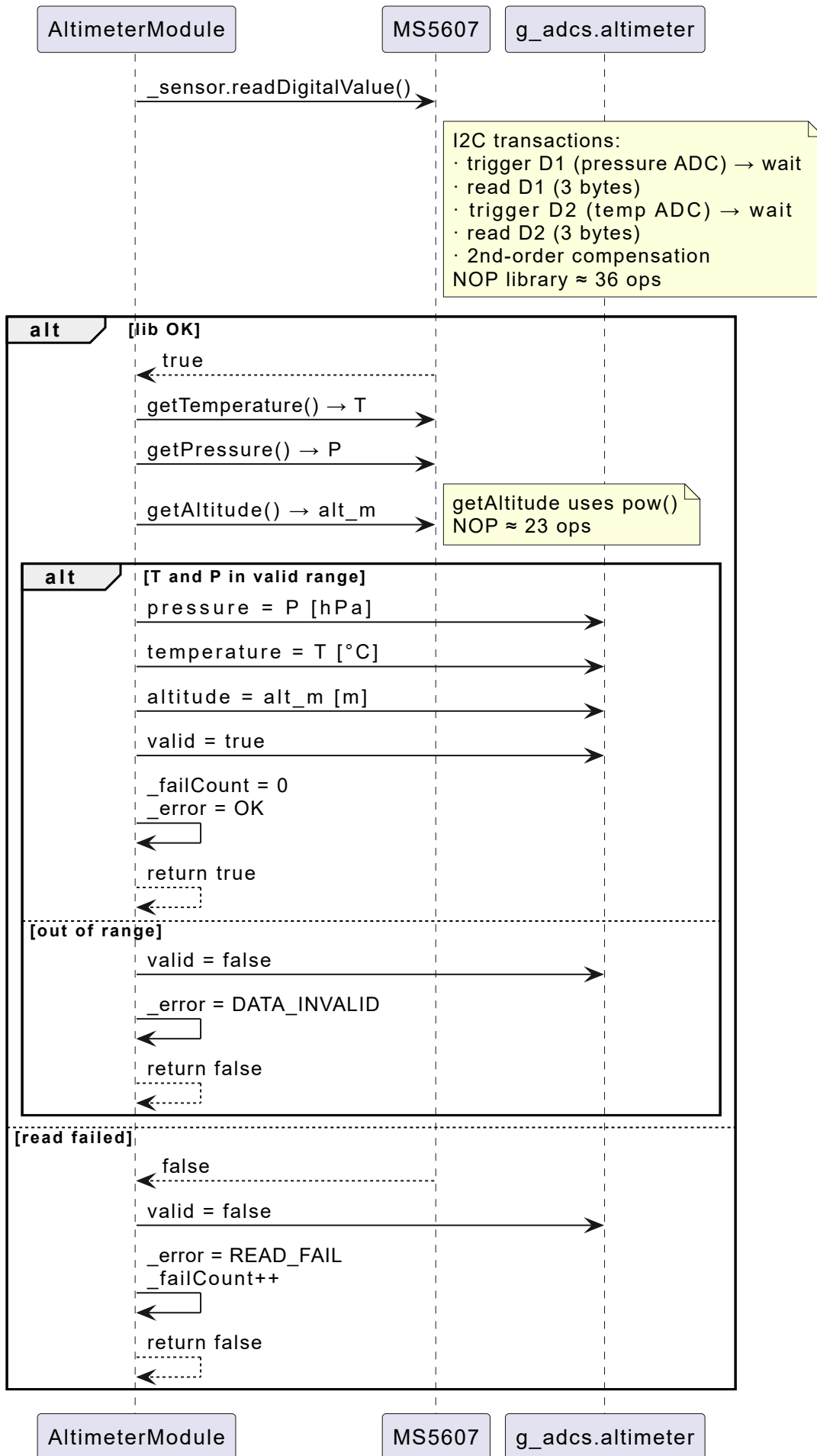
2. Class Diagram



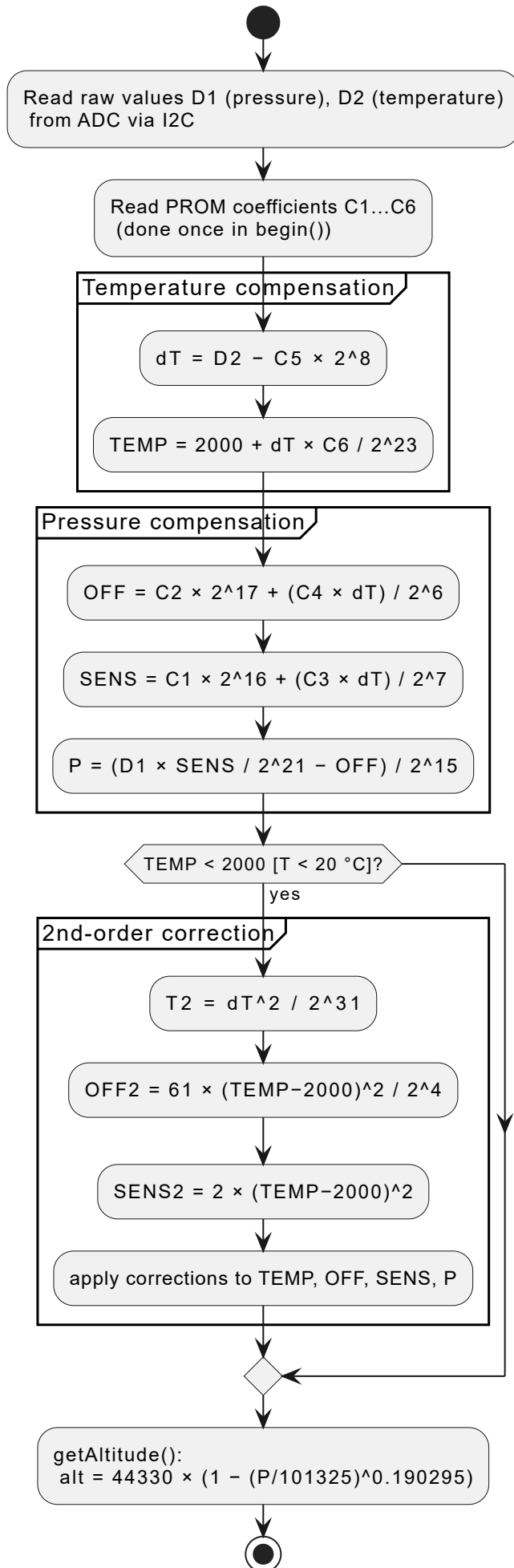
3. Initialization Sequence — `init()`



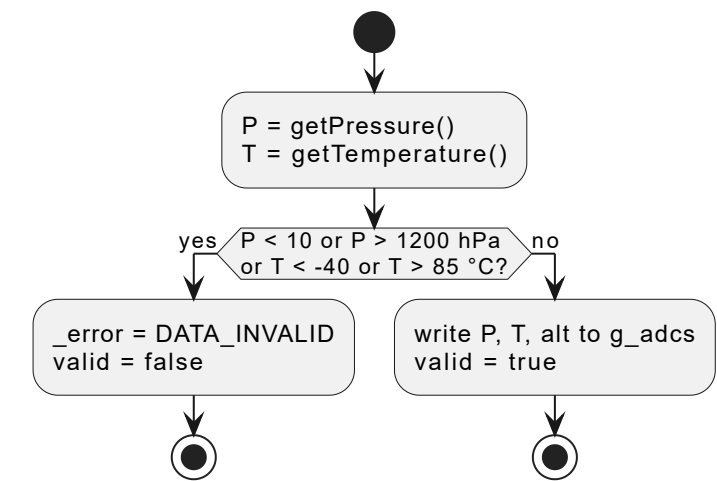
4. Measurement Sequence — `read()`



5. MS5607 Compensation Formulas



6. Validation Ranges (firmware-enforced)



Field	Valid range
Pressure	10 – 1200 hPa
Temperature	–40 °C – +85 °C
Altitude	derived — not independently validated

7. Output — `g_adcs.altimeter`

Field	Unit	Source
<code>pressure</code>	hPa	MS5607 ADC + compensation
<code>temperature</code>	°C	MS5607 ADC + compensation
<code>altitude</code>	m	ISA barometric formula
<code>valid</code>	bool	false on read fail or out-of-range